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Department: AI-DS(A)

Subject: Vsec

# Assignment No. 06

**1. System Preparation**

**Step 1.1 — Update package index**

sudo apt update

**Purpose:** Refresh package metadata
**Failure case:**

- Temporary failure resolving → network/DNS issue

---

**Step 1.2 — Upgrade packages**

sudo apt upgrade -y

**Purpose:** Ensure compatibility with Docker dependencies
**Note:** May take time

---

**2. Validate Docker Runtime**

**Step 2.1 — Check Docker version**

docker --version

**Expected:**

Docker version XX.X.X

**Failure:**

- command not found → Docker not installed

---

**Step 2.2 — Check Docker service**

sudo systemctl status docker

**Expected:**

- active (running)

**Exit screen:**

q

**Failure cases:**

- inactive → start manually:

sudo systemctl start docker

---

**Step 2.3 — Test Docker with sample container**

docker run hello-world

**Expected output:**

- Confirmation message from Docker

**Failure modes:**

- Permission denied → fix:

sudo usermod -aG docker $USER
newgrp docker

---

**Step 2.4 — Check Docker Compose**

docker compose version

**Failure:**

sudo apt install docker-compose-plugin

---

**Step 2.5 — Inspect Docker environment**

docker info

**Check:**

- Storage driver
- CPU/memory detected

---

**3. Project Setup**

**Step 3.1 — Create project directory**

mkdir ai_docker_project

**Step 3.2 — Enter directory**

cd ai_docker_project

---

**4. Create AI Model Script**

**Step 4.1 — Create file**

nano model.py

Paste:

```
import numpy as np
print("AI Model Output:", np.mean([10, 20, 30, 40]))
```

Save: CTRL + X → Y → ENTER

---

**Step 4.2 — Test locally (critical validation)**

python3 model.py

**Expected:**

AI Model Output: 25.0

**Why this matters:**

- Confirms Python + NumPy logic works before containerization

---

**5. Define Dependencies**

**Step 5.1 — Create requirements file**

nano requirements.txt

Paste:

numpy

Save and exit.

---

**6. Create Dockerfile**

**Step 6.1 — Create file**

nano Dockerfile

Paste:

```
FROM python:3.10-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY model.py .

CMD ["python", "model.py"]
```

---

**7. Build Docker Image**

**Step 7.1 — Build image**

```
docker build -t ai-model-container .
```

---

**Step 7.2 — Verify image**

```
docker images
```

**Expected:**

```
ai-model-container   latest   <image_id>
```

---

**8. Run Container**

**Step 8.1 — Execute container**

```
docker run ai-model-container
```

**Expected output:**

```
AI Model Output: 25.0
```

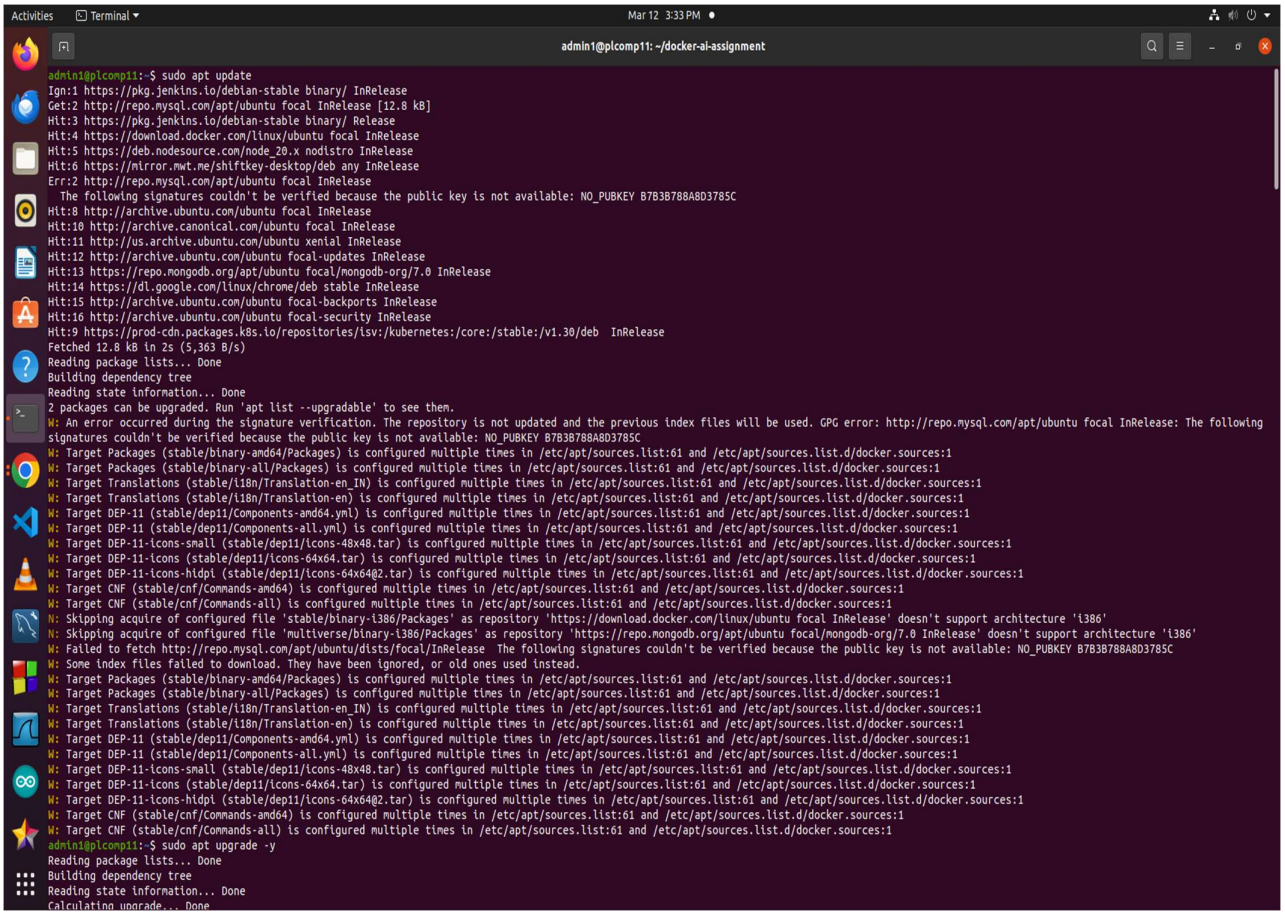
## 9. Inspect Containers

### Step 9.1 — List all containers

docker ps -a

**Expected:**

- Container listed with status Exited



Activities  Terminal  Mar 12  3:37 PM

admin1@plcomp11: ~/docker-ai-assignment

```
  libglib2.0-0 bind9-libs librados2 libcurl3-gnutls gpgconf gstreamer1.0-gl
  git-cvs python3-apt libtiffxx5 libctf0 cups-browsed libgcc-9-dev libxslt1.1
  git-doc gpgsm git-el libkpathsea6 libpng16-16
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following packages have been kept back:
  mysql-client mysql-server
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
admin1@plcomp11:~$ docker --version
Docker version 28.1.1, build 4eba377
admin1@plcomp11:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset>
     Active: active (running) since Thu 2026-03-12 15:01:34 IST; 19min ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2550 (dockerd)
      Tasks: 20
     Memory: 100.3M
     CGroup: /system.slice/docker.service
             └─2550 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/cont>

Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.705183123+05:>
Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.735146353+05:>
Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.920553296+05:>
Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.943353316+05:>
Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.959361553+05:>
Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.959581726+05:>
Mar 12 15:01:34 plcomp11 dockerd[2550]: time="2026-03-12T15:01:34.039330013+05:>
Mar 12 15:01:34 plcomp11 dockerd[2550]: time="2026-03-12T15:01:34.044222561+05:>
Mar 12 15:01:34 plcomp11 dockerd[2550]: time="2026-03-12T15:01:34.044251242+05:>
Mar 12 15:01:34 plcomp11 systemd[1]: Started Docker Application Container Engin>
lines 1-21/21 (END)...skipping...
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset>
     Active: active (running) since Thu 2026-03-12 15:01:34 IST; 19min ago
TriggeredBy: ● docker.socket
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      Tasks: 20
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Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.705183123+05:>
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Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.920553296+05:>
Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.943353316+05:>
Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.959361553+05:>
Mar 12 15:01:33 plcomp11 dockerd[2550]: time="2026-03-12T15:01:33.959581726+05:>
Mar 12 15:01:34 plcomp11 dockerd[2550]: time="2026-03-12T15:01:34.039330013+05:>
Mar 12 15:01:34 plcomp11 dockerd[2550]: time="2026-03-12T15:01:34.044222561+05:>
Mar 12 15:01:34 plcomp11 dockerd[2550]: time="2026-03-12T15:01:34.044251242+05:>
Mar 12 15:01:34 plcomp11 systemd[1]: Started Docker Application Container Engin>
~
~
~
```

Activities  Terminal  Mar 12  3:37 PM

admin1@plcomp11: ~/docker-ai-assignment

```
~
~
~
admin1@plcomp11:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:85404b3c53951c3ff5d40de0972b1bb21fafa2e8daa235355baf44f33db9dbdd
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

admin1@plcomp11:~$ docker compose version
Docker Compose version v5.0.1
admin1@plcomp11:~$ docker info
Client: Docker Engine - Community
 Version:    28.1.1
 Context:    default
 Debug Mode: false
 Plugins:
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.17.1
    Path:     /home/admin1/.docker/cli-plugins/docker-ai
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.30.1-desktop.2
    Path:     /home/admin1/.docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.0.1
    Path:     /home/admin1/.docker/cli-plugins/docker-compose
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     /home/admin1/.docker/cli-plugins/docker-debug
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.2.0
    Path:     /home/admin1/.docker/cli-plugins/docker-desktop
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
```

Activities  Terminal  Mar 12  3:45 PM

admin1@plcomp11: ~/ai_docker_project

```
admin1@plcomp11:~$ mkdir ai_docker_project
admin1@plcomp11:~$ cd ai_docker_project
admin1@plcomp11:~/ai_docker_project$ nano model.py
admin1@plcomp11:~/ai_docker_project$ python3 model.py
AI Model Output: 25.0
admin1@plcomp11:~/ai_docker_project$ nano requirements.txt
admin1@plcomp11:~/ai_docker_project$ nano Dockerfile
admin1@plcomp11:~/ai_docker_project$ docker build -t ai-model-container .
[+] Building 20.0s (10/10) FINISHED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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=> [internal] load build definition from Dockerfile                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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=> => transferring dockerfile: 196B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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=> [internal] load metadata for docker.io/library/python:3.10-slim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       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=> [internal] load .dockerignore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         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=> => transferring context: 2B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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=> [1/5] FROM docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aab281d705d224457                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                5.9s
=> => resolve docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aab281d705d224457                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419 29.78MB / 29.78MB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                4.8s
=> => sha256:aecf3e6b3e04ba467f457952ea77eb6cebfb10ea3231a822d131ceca37129d60 1.29MB / 1.29MB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.3s
=> => sha256:44df7c9b01189cee78fe34391e9a58e4a1d449a932b41113d54fa1c9b03f828b 13.82MB / 13.82MB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                4.9s
=> => sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aab281d705d224457 10.37kB / 10.37kB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => sha256:1dc3b6808fb4739068071dc9543150ecc87be29f4f8af018c47af841578804ff 1.75kB / 1.75kB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => sha256:c2cb0450ef763646d50b7b6ed8ce95d3f52edeb354e57d62c2169a67553b0127 5.48kB / 5.48kB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => sha256:7fe6b279891367efd009a54b83403e92c54fe9b9e47c0e5e468533209b2ad068 252B / 252B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                3.2s
=> => extracting sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.4s
=> => extracting sha256:aecf3e6b3e04ba467f457952ea77eb6cebfb10ea3231a822d131ceca37129d60                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => extracting sha256:44df7c9b01189cee78fe34391e9a58e4a1d449a932b41113d54fa1c9b03f828b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.2s
=> => extracting sha256:7fe6b279891367efd009a54b83403e92c54fe9b9e47c0e5e468533209b2ad068                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> [internal] load build context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => transferring context: 159B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> [2/5] WORKDIR /app                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.2s
=> [3/5] COPY requirements.txt .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                8.5s
=> [5/5] COPY model.py .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> exporting to image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.5s
=> => exporting layers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.5s
=> => writing image sha256:e00fcd61aee628b64679958a244a746cdc74d37397fdc76eb5a85cc55582fa1b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => naming to docker.io/library/ai-model-container                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
admin1@plcomp11:~/ai_docker_project$ docker images
REPOSITORY                  TAG                 IMAGE ID       CREATED          SIZE
ai-model-container          latest              e00fcd61aee6   11 seconds ago   195MB
ai-model                    latest              1d3280890f00   47 hours ago     1.1GB
my-ai-app                   latest              741af8270ef2   2 days ago       1.2GB
gcr.io/k8s-minikube/kicbase v0.0.50             6da180ef5035   3 weeks ago      1.37GB
hello-world                 latest              1b44b5a3e06a   7 months ago     10.1kB
apache/spark                3.5.0               b7bd251f1d58   2 years ago      981MB
bde2020/hadoop-nodemanager  2.0.0-hadoop3.2.1-java8  4e47dabd148f  6 years ago  1.37GB
bde2020/hadoop-resourcemanager  2.0.0-hadoop3.2.1-java8  3deba4a1885f  6 years ago  1.37GB
bde2020/hadoop-namenode     2.0.0-hadoop3.2.1-java8  839ec11d95f8  6 years ago  1.37GB
bde2020/hadoop-datanode     2.0.0-hadoop3.2.1-java8  df288ee0a7f9  6 years ago  1.37GB
admin1@plcomp11:~/ai_docker_project$ docker run ai-model-container
AI Model Output: 25.0
admin1@plcomp11:~/ai_docker_project$ docker ps -a
CONTAINER ID   IMAGE                                          COMMAND                  CREATED          STATUS                     PORTS     NAMES
8b60049ffff7   ai-model-container                             "python model.py"        39 seconds ago   Exited (0) 38 seconds ago            agitated_mirzakhani
7bd6d6c49a80   hello-world                                    "/hello"                 18 minutes ago   Exited (0) 18 minutes ago            relaxed_neumann
61901b660184   bde2020/hadoop-nodemanager:2.0.0-hadoop3.2.1-java8  "/entrypoint.sh /run…"  42 hours ago  Exited (137) 41 hours ago            nodemanager
```


Activities  Terminal  Mar 12  3:45 PM

admin1@plcomp11: ~/ai_docker_project

```
[+] Building 20.0s (10/10) FINISHED                                                                                                                                                docker:default
=> [internal] load build definition from Dockerfile                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => transferring dockerfile: 196B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> [internal] load metadata for docker.io/library/python:3.10-slim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                4.4s
=> [internal] load .dockerignore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => transferring context: 2B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> [1/5] FROM docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aab281d705d224457                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                5.9s
=> => resolve docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aab281d705d224457                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419 29.78MB / 29.78MB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                4.8s
=> => sha256:aecf3e6b3e04ba467f457952ea77eb6cebfb10ea3231a822d131ceca37129d60 1.29MB / 1.29MB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.3s
=> => sha256:44df7c9b01189cee78fe34391e9a58e4a1d449a932b41113d54fa1c9b03f828b 13.82MB / 13.82MB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                4.9s
=> => sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aab281d705d224457 10.37kB / 10.37kB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => sha256:1dc3b6808fb4739068071dc9543150ecc87be29f4f8af018c47af841578804ff 1.75kB / 1.75kB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => sha256:c2cb0450ef763646d50b7b6ed8ce95d3f52edeb354e57d62c2169a67553b0127 5.48kB / 5.48kB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => sha256:7fe6b279891367efd009a54b83403e92c54fe9b9e47c0e5e468533209b2ad068 252B / 252B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                3.2s
=> => extracting sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.4s
=> => extracting sha256:aecf3e6b3e04ba467f457952ea77eb6cebfb10ea3231a822d131ceca37129d60                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => extracting sha256:44df7c9b01189cee78fe34391e9a58e4a1d449a932b41113d54fa1c9b03f828b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.2s
=> => extracting sha256:7fe6b279891367efd009a54b83403e92c54fe9b9e47c0e5e468533209b2ad068                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> [internal] load build context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => transferring context: 159B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> [2/5] WORKDIR /app                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.2s
=> [3/5] COPY requirements.txt .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                8.5s
=> [5/5] COPY model.py .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> exporting to image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.5s
=> => exporting layers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.5s
=> => writing image sha256:e00fcd61aee628b64679958a244a746cdc74d37397fdc76eb5a85cc55582fa1b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => naming to docker.io/library/ai-model-container                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```


```
admin1@plcomp11:~/ai_docker_project$ docker images
REPOSITORY                  TAG                 IMAGE ID       CREATED          SIZE
ai-model-container          latest              e00fcd61aee6   11 seconds ago   195MB
ai-model                    latest              1d3280890f00   47 hours ago     1.1GB
my-ai-app                   latest              741af8270ef2   2 days ago       1.2GB
gcr.io/k8s-minikube/kicbase v0.0.50             6da180ef5035   3 weeks ago      1.37GB
hello-world                 latest              1b44b5a3e06a   7 months ago     10.1kB
apache/spark                3.5.0               b7bd251f1d58   2 years ago      981MB
bde2020/hadoop-nodemanager  2.0.0-hadoop3.2.1-java8  4e47dabd148f  6 years ago  1.37GB
bde2020/hadoop-resourcemanager  2.0.0-hadoop3.2.1-java8  3deba4a1885f  6 years ago  1.37GB
bde2020/hadoop-namenode     2.0.0-hadoop3.2.1-java8  839ec11d95f8  6 years ago  1.37GB
bde2020/hadoop-datanode     2.0.0-hadoop3.2.1-java8  df288ee0a7f9  6 years ago  1.37GB
admin1@plcomp11:~/ai_docker_project$ docker run ai-model-container
AI Model Output: 25.0
admin1@plcomp11:~/ai_docker_project$ docker ps -a
CONTAINER ID   IMAGE                                          COMMAND                  CREATED          STATUS                     PORTS     NAMES
8b60049ffff7   ai-model-container                             "python model.py"        39 seconds ago   Exited (0) 38 seconds ago            agitated_mirzakhani
7bd6d6c49a80   hello-world                                    "/hello"                 18 minutes ago   Exited (0) 18 minutes ago            relaxed_neumann
61901b660184   bde2020/hadoop-nodemanager:2.0.0-hadoop3.2.1-java8  "/entrypoint.sh /run…"  42 hours ago  Exited (137) 41 hours ago            nodemanager
840576f27e57   bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8  "/entrypoint.sh /run…"  42 hours ago  Exited (137) 41 hours ago            datanode
f896d63cb538   bde2020/hadoop-resourcemanager:2.0.0-hadoop3.2.1-java8  "/entrypoint.sh /run…"  42 hours ago  Exited (137) 41 hours ago            resourcemanager
08ab1694791b   bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8  "/entrypoint.sh /run…"  42 hours ago  Exited (137) 41 hours ago            namenode
5fa470d609c8   apache/spark:3.5.0                             "/opt/entrypoint.sh"     42 hours ago     Exited (0) 42 hours ago              spark
d979aeb27791   gcr.io/k8s-minikube/kicbase:v0.0.50            "/usr/local/bin/entr…"   2 days ago       Exited (137) 47 hours ago            minikube
82c1cacdc2e3   my-ai-app                                      "python app.py"          2 days ago       Exited (137) 47 hours ago            confident_fermi
admin1@plcomp11:~/ai_docker_project$
```